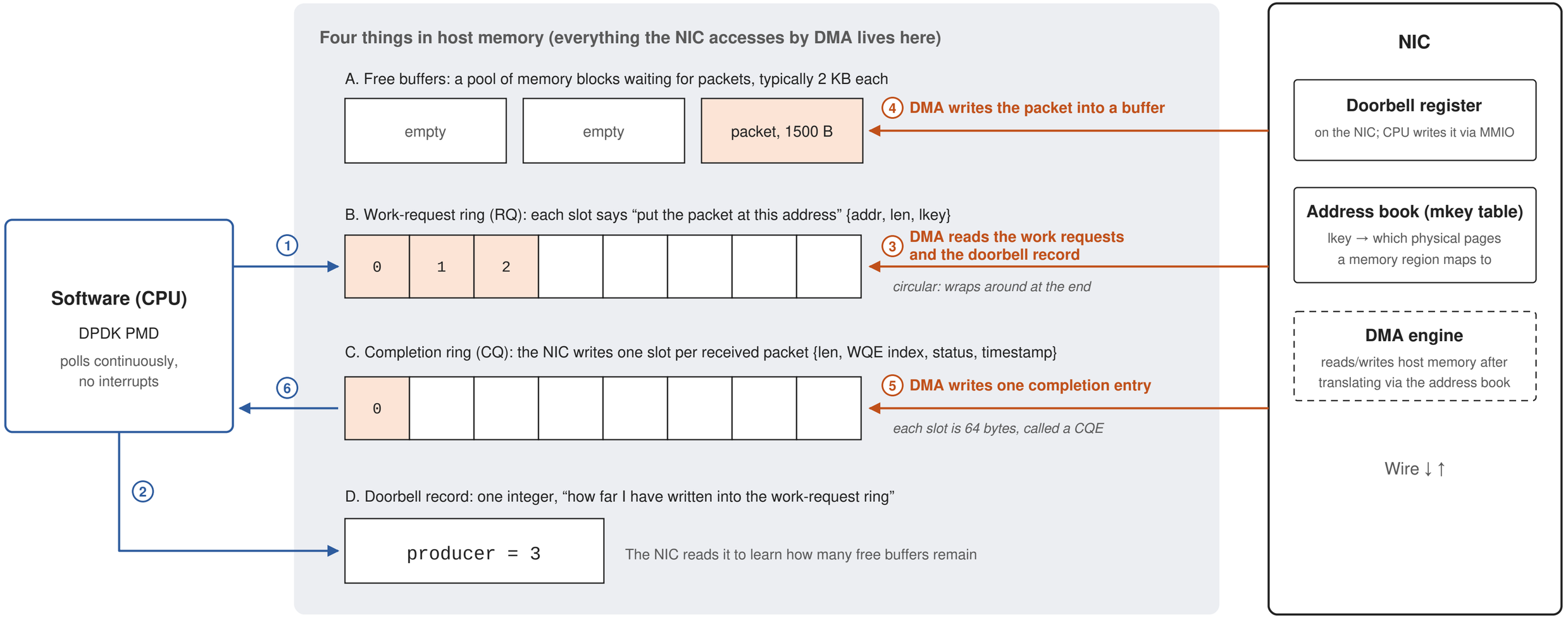
Four things in host memory (everything the NIC accesses by DMA lives here)
A. Free buffers: a pool of memory blocks waiting for packets, typically 2 KB each
empty
empty
packet, 1500 B
4 DMA writes the packet into a buffer
B. Work-request ring (RQ): each slot says “put the packet at this address” {addr, len, lkey}
0
1
2
3 DMA reads the work requests and the doorbell record
circular: wraps around at the end
C. Completion ring (CQ): the NIC writes one slot per received packet {len, WQE index, status, timestamp}
0
5 DMA writes one completion entry
each slot is 64 bytes, called a CQE
D. Doorbell record: one integer, “how far I have written into the work-request ring”
producer = 3
The NIC reads it to learn how many free buffers remain
Software (CPU)
DPDK PMD
polls continuously, no interrupts
1
2
6
NIC
Doorbell register
on the NIC; CPU writes it via MMIO
Address book (mkey table)
lkey → which physical pages a memory region maps to
DMA engine
reads/writes host memory after translating via the address book
Wire ↓ ↑